

Cloud Computing

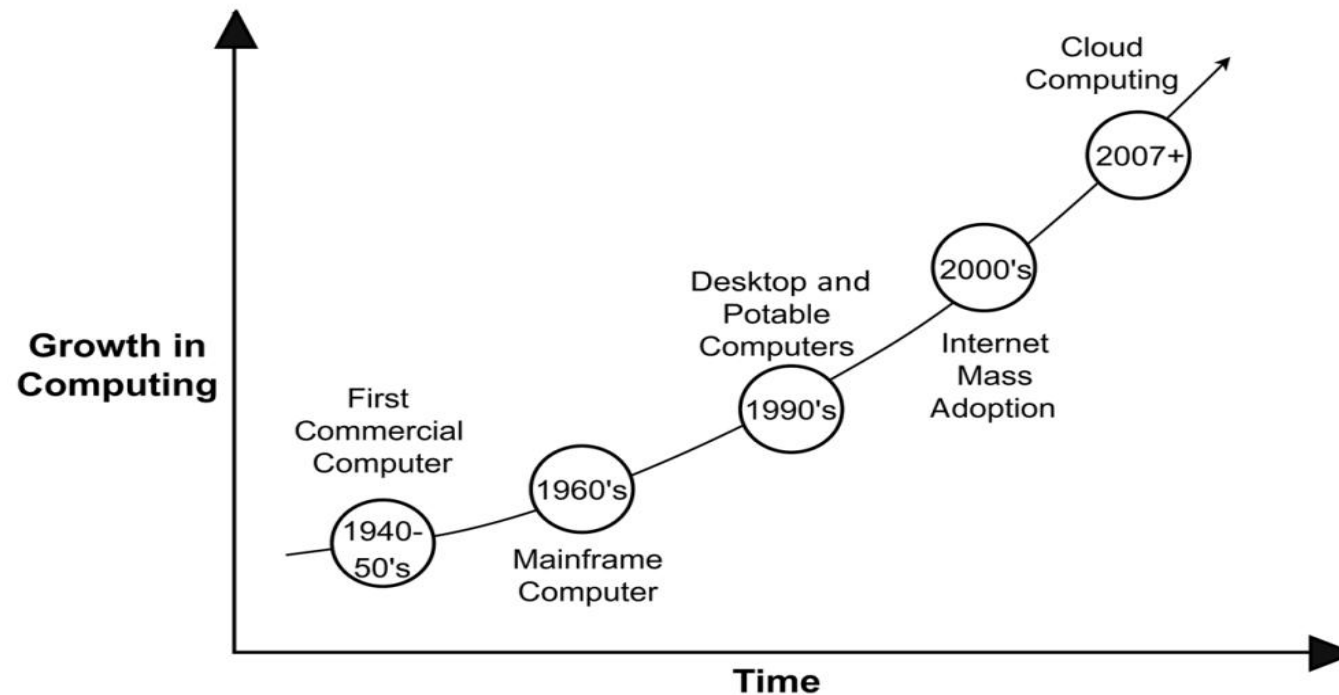
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Cloud Analytics for IoT Applications

- ▶ Cloud analytics for IoT applications refers to the use of cloud computing platforms and data analytics techniques to process, analyze, and gain insights from the vast amount of data generated by Internet of Things (IoT) devices.

Introduction to Cloud Computing

- Cloud computing is a technological advancement that enables the distribution of computing services via the internet, encompassing servers, storage, databases, networking, software, analytics, and intelligence, among other resources.



Cloud Computing

- ▶ In cloud computing, "cloud" refers to a network of remote servers hosted on the internet, rather than being stored on local servers or personal computers. These servers provide various services such as storage, computing power, databases, and software, which can be accessed over the internet.
- ▶ So, the cloud is a metaphor for the internet, but specifically it refers to the infrastructure and services provided via the internet.
- ▶ It essentially provides computing resources as a service, whereby third-party providers possess and operate the resources for the benefit of the end-users.

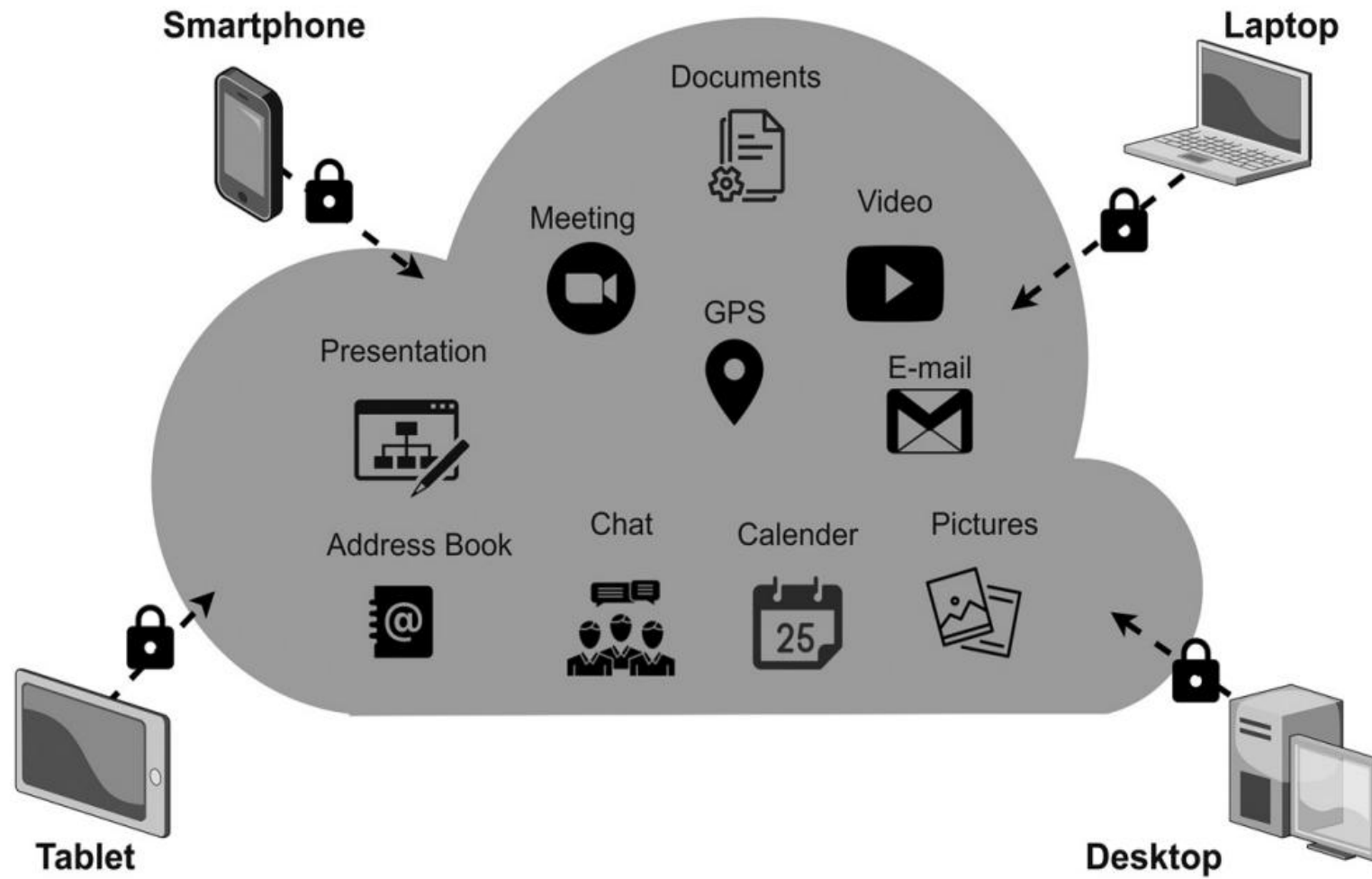
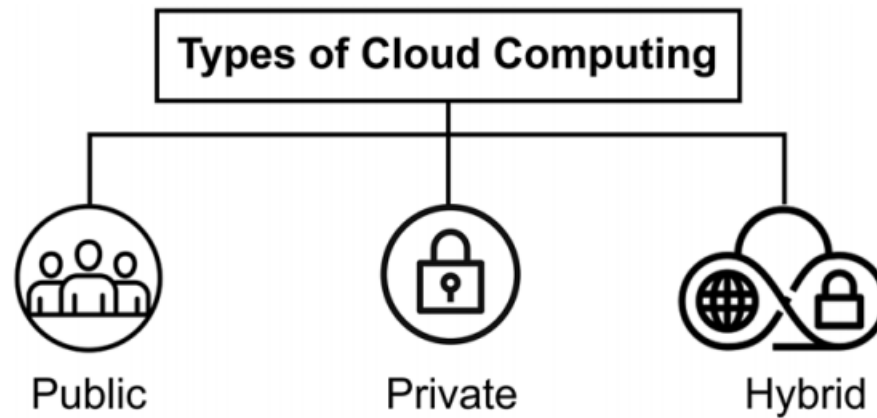


FIGURE 5.2 Application of the cloud.

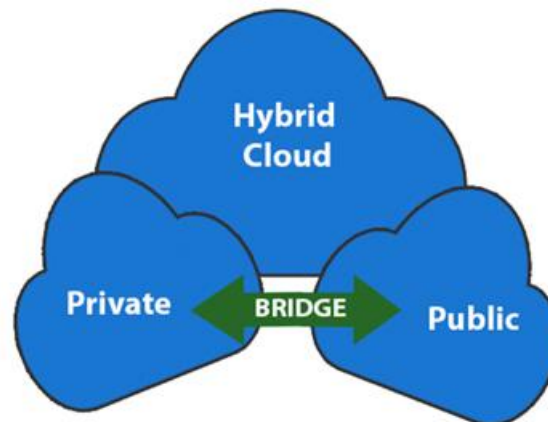
Categories of Cloud Computing

Public Cloud: Public clouds refer to cloud resources that are owned and operated by a third-party cloud service provider. Users access computing resources, such as servers, software, and storage, through the internet. Public clouds are open to multiple users.



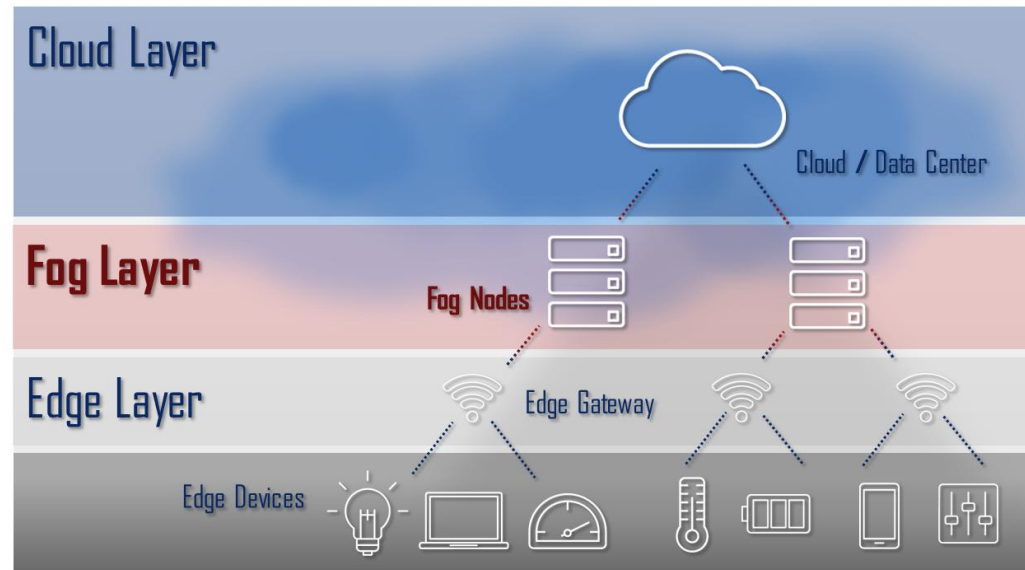
Private Cloud: Private clouds are exclusive to a single business or organization. These resources may be located on-site or hosted by a third-party. Private clouds are typically more secure than public clouds, as access is restricted to authorized personnel only.

Hybrid Cloud: Hybrid clouds are a combination of public and private clouds. Data and applications are shared between the two, allowing for greater flexibility and deployment options. Hybrid clouds allow businesses to take advantage of the scalability and cost effectiveness of public clouds, while also ensuring the security and compliance of private clouds.



Fog Computing

- ▶ Fog Computing is a distributed computing model where data processing happens closer to the data source (near IoT devices) instead of sending everything to the cloud.
- ▶ Fog computing creates a link between cloud computing and the internet of things, making it possible to process data locally across various nodes.



Fog vs Cloud Computing

Architecture: Cloud computing architecture is centralized, with data processed in large data centers, while fog computing architecture is decentralized, with data processed at the edge of the network.

Processing: In cloud computing, data is processed in a remote data center, while in fog computing, data is processed locally on multiple edge nodes.

Latency: Cloud computing has higher latency as it involves transferring data to remote data centers, while fog computing has lower latency due to the local processing of data.

Bandwidth: In cloud computing, large amounts of bandwidth are required to transmit data to remote data centers, while in fog computing, less bandwidth is required as data is processed locally.

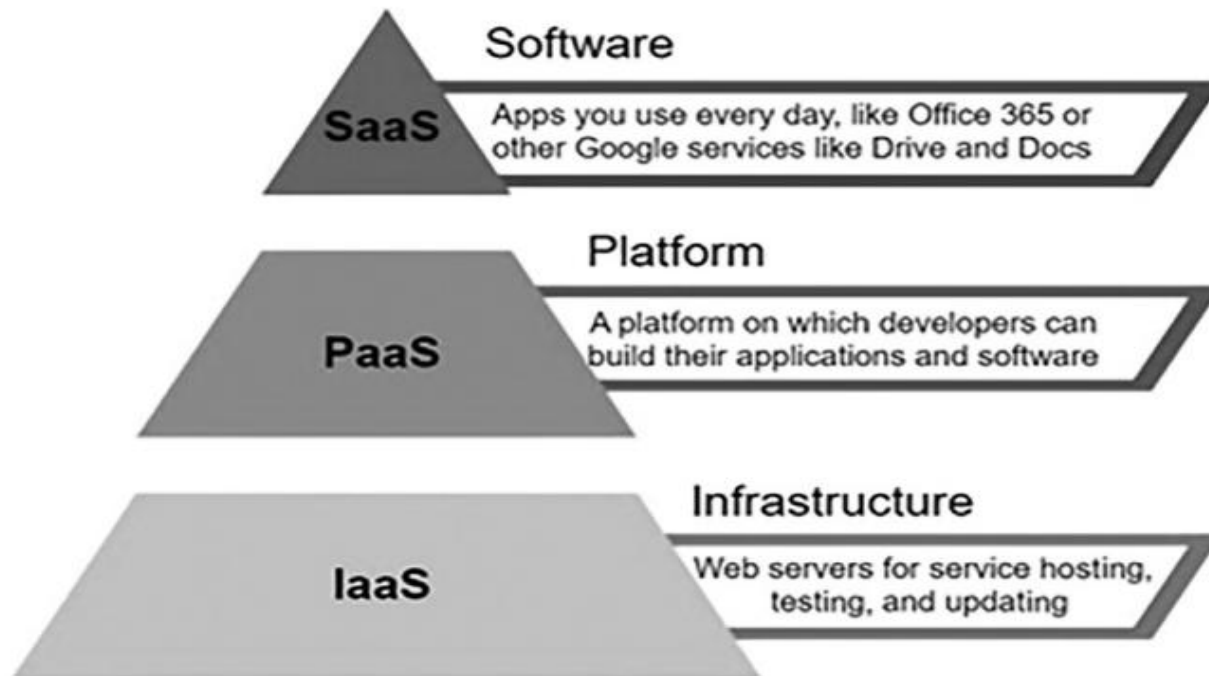
The background of the slide features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, ranging from light lime to dark forest green. These shapes are concentrated on the right side of the slide, creating a modern, tech-oriented aesthetic.

Security: Cloud computing offers a high level of security through data encryption and other security measures, while fog computing offers additional security measures due to its decentralized architecture.

Use cases: Cloud computing is ideal for applications that require significant computational power, storage, and data processing, while fog computing is better suited for applications that require real-time processing, low latency, and data proximity.

Evolution of Cloud Computing

- The next evolution of cloud computing centers around delivering software, platforms, and infrastructure as services, commonly known as SaaS, PaaS, and IaaS.



Software-as-a-Service (SaaS): SaaS is a model in which a cloud provider hosts a software application and offers it to customers over the internet. It includes email services, office applications, and customer relationship management (CRM) software.

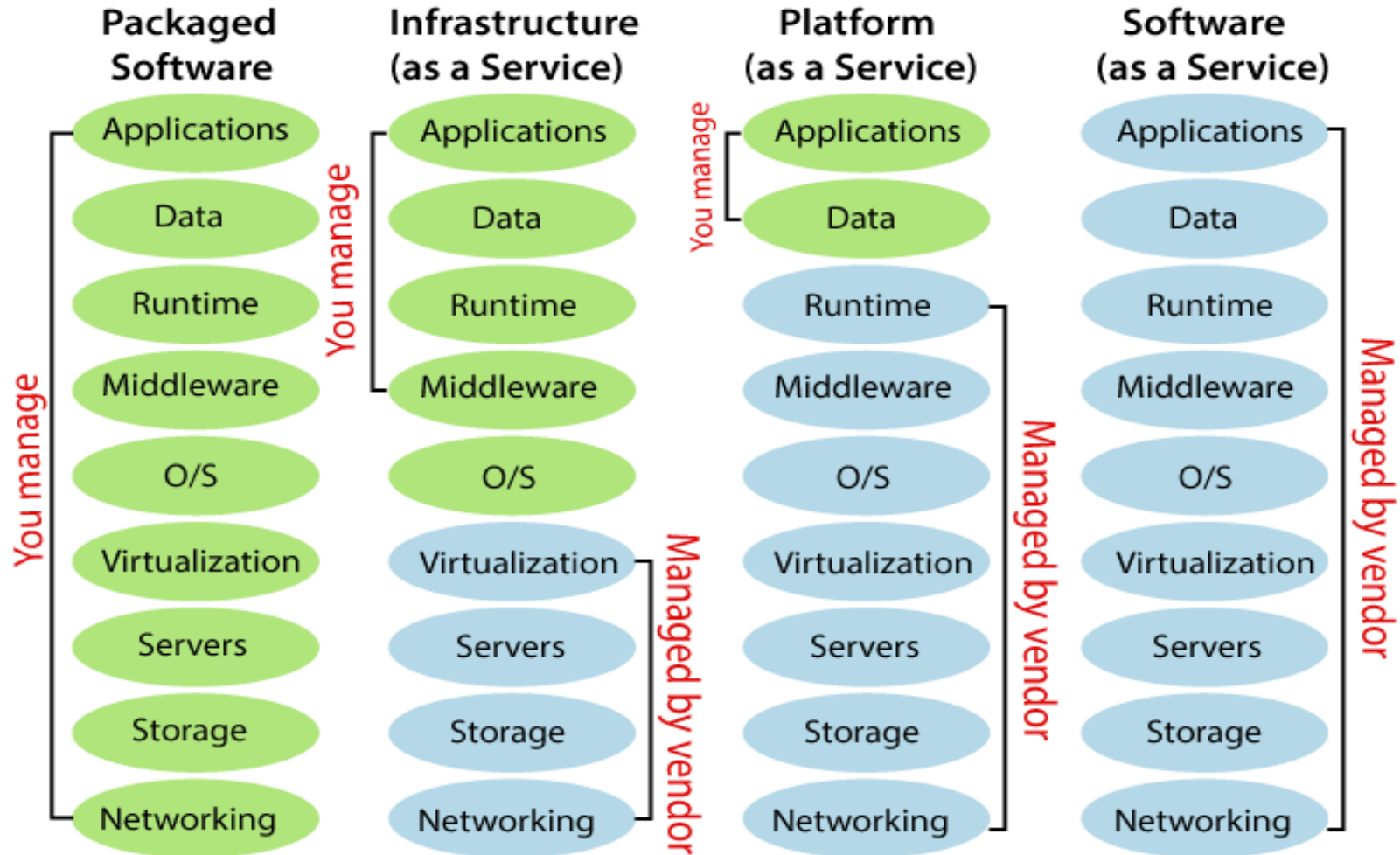
Platform-as-a-Service (PaaS): PaaS is a type of cloud service that enables customers to develop, operate, and manage applications without being concerned about the underlying infrastructure. The cloud provider manages the infrastructure, operating system, and middleware, allowing the customers to concentrate on the development and deployment of their applications. Google App Engine and Microsoft Azure are examples of PaaS.

Infrastructure-as-a-Service (IaaS): IaaS is a cloud computing model that enables users to access and manage computing resources such as servers, storage, and networking over the internet. It allows businesses to build their own IT infrastructure without the need to invest in or maintain physical hardware. Popular examples of IaaS providers include Amazon Web Services (AWS) and Microsoft Azure.

IaaS → you control virtual machine (AWS)

PaaS → ready coding platform (Google Colab)

SaaS → finished application (Google Docs)



Advantages of Cloud Computing

- ▶ **Cost:** It reduces the huge capital costs of buying hardware and software.
- ▶ **Speed:** Resources can be accessed in minutes, typically within a few clicks.
- ▶ **Scalability:** We can increase or decrease the requirement of resources according to the business requirements.
- ▶ **Productivity:** While using cloud computing, we put less operational effort. We do not need to apply patching, as well as no need to maintain hardware and software. So, in this way, the IT team can be more productive and focus on achieving business goals.
- ▶ **Reliability:** Backup and recovery of data are less expensive and very fast for business continuity.
- ▶ **Security:** Many cloud vendors offer a broad set of policies, technologies, and controls that strengthen our data security.

Role of Cloud Computing in IoT

- ▶ The term “cloud” denotes a centralized framework that enables the transmission and distribution of data and files to distant data centers via the internet.
- ▶ By integrating IoT with cloud computing, innovative solutions have been developed that provide valuable insights into intricate systems and improve operational efficiency.
- ▶ For example, cloud computing can be utilized to store and upload sensor data for future smart monitoring and activation with other devices.
- ▶ The primary aim is to transform data into insights that can enhance profitable and productive operations.

Benefits

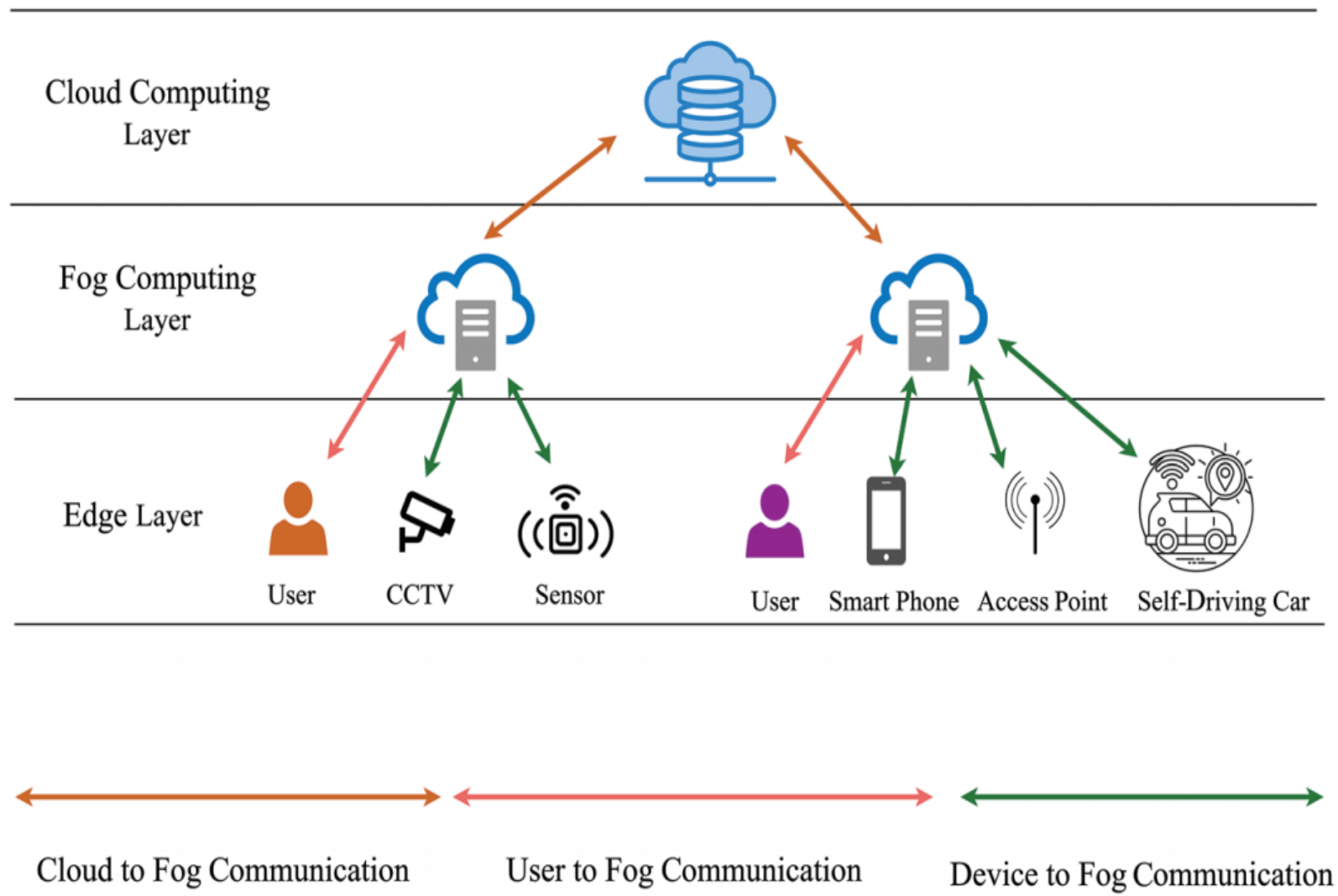
- ▶ According to AWS, cloud computing has the following four main benefits:
 - It eliminates the need to guess infrastructure capacity needs in advance.
 - It allows for cost savings by only paying for the resources that are used, resulting in greater savings as volume increases.
 - The platform can be deployed worldwide within minutes.
 - It offers developers the advantage of fast and flexible resource allocation.

When to use Fog Computing?

1. Selective data transmission to the cloud, whereby only specific data is transmitted for long-term storage and accessed less frequently by the host.
2. The need for multiple services across different geographic locations, especially in a wide area.
3. Devices that require extensive computations and processing can benefit from fog computing.
4. Fog computing is used in real-world examples such as IoT devices, the IIoT that includes devices with sensors and cameras, and the CAR 2 CAR Communication Consortium.

Drawbacks of Fog Computing

1. Authentication: Fog computing involves a large number of interconnected devices, which makes it challenging to authenticate each device due to its distributed nature.
2. Scheduling: Moving tasks between fog devices, fog nodes, and cloud servers creates scheduling issues.
3. Power Consumption: Fog nodes consume more power than centralized cloud architecture, which can be a concern.
4. Fog Servers: Proper placement of fog servers is necessary to ensure optimal service delivery.



Edge Computing

- ▶ Edge computing refers to the practice of bringing computing resources closer to the data source or “edge” of a network.
- ▶ By processing data at the edge, edge computing enables real-time or near-real-time analysis, faster response times, and reduced network congestion.
- ▶ Edge computing refers to a decentralized computing paradigm that brings computation and data storage closer to the edge of the network—closer to the data source or end-user devices.

Challenges in Edge Computing

- Privacy and Security of Data: Edge computing requires advanced security schemes to ensure the privacy and security of data. Different encryption
- Scalability: The distributed network structure of edge computing makes scalability challenging. The heterogeneity of devices and their varying energy and performance constraints, as well as dynamic conditions and reliability of connections, create several issues.
 - Reliability: Ensuring the dependability of service delivery is critical in edge computing. It is important for each device to maintain the network topology of the entire distributed system to identify errors and offer recovery measures.
- Speed: Edge computing is expected to deliver speedy services to end-users by utilizing analytical and computational resources in close proximity to the source.
- Efficiency: Edge computing becomes more efficient when advanced analytical tools are located closer to end-users.

Edge Computing

Edge computing is ideal for organizations with limited financial resources, such as mid-level organizations.

Edge computing enables the utilization of various programming languages on diverse platforms, each having a unique runtime.

Edge computing is responsible for handling time-critical information by performing data processing tasks at the edge of the network.

Edge computing employs a distributed model and performs data processing at remote locations. It follows a decentralized approach, enabling it to operate closer to the source of the data it is processing.

Edge computing enables organizations to utilize their current IoT devices and enhance their capabilities without having to buy new devices.

Edge computing is the future of computing.

Cloud Computing

Cloud computing is recommended for managing and processing a high volume of complex and massive data. Therefore, organizations dealing with massive data storage use cloud computing.

Cloud computing operates on a particular platform and employs a specific programming language.

Cloud computing is a type of data processing that is not constrained by time. This means that the data processing is not dependent on specific timeframes.

Cloud computing adopts a centralized approach to manage and process data at centralized locations.

Cloud computing requires the use of new IoT devices to replace existing ones, which is a costly and time-consuming process.

Cloud computing is the currently existing technology.